

CHE 565 -- Homework 3

Soki Sem

April 18, 2026

Problem 1

Steepest Descent

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At $x_0 = \text{Matrix}([1, 1])$, $f(x_0) = 0.5000000000000000$

Gradient of f:

$$\nabla f = \begin{bmatrix} 1.0x_1 + 1.0x_2 - 1.0 \\ 1.0x_1 + 2.0x_2 - 1 \end{bmatrix}$$

At $x_0 = \text{Matrix}([1, 1])$, $Df(x_0) = 1.00, 2.00$

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 - 1.0\alpha \\ 1 - 2.0\alpha \end{bmatrix}$$

Function ϕ_k :

$$\phi(\alpha) = 6.5\alpha^2 - 5.0\alpha + 0.5$$

Derivative of ϕ_k :

$$\begin{aligned} \frac{d}{d\alpha}\phi(\alpha) &= 13.0\alpha - 5.0 \\ 0 &= 13.0\alpha - 5.0 \end{aligned}$$

Optimal alpha: 0.38

Next iterate x_1 : 0.62, x_2 : 0.23

Function value at next iterate: -0.46

Problem 2

Conjugate Gradient Method

$$\begin{bmatrix} s_1(\alpha) \\ s_2(\alpha) \end{bmatrix} = \begin{bmatrix} 0.0946745562130178 \\ -0.195266272189349 \end{bmatrix}$$

Problem 3

Constituents	Maximum Quantity (bbl/day)	Production Cost (\$/bbl)
1	3000	26.00
2	2000	30.60
3	4000	29.20
4	1000	29.80

Grade	Specifications	Selling Price (\$/bbl)
A	No more than 15% of 1; No more than 40% of 2	32.40
B	No more than 50% of 3; No more than 10% of 1	31.50
C	No less than 10% of 2; No more than 20% of 1	30.60

$$[a_1 \leq 0.15A, a_2 \geq 0.4A, a_3 \leq 0.5A, b_1 \leq 0.1B, b_2 \geq 0.1B, c_1 \leq 0.2C, a_1 + b_1 + c_1 \leq 3000, a_2 + b_2 + c_2 \leq 2000, a_3 + b_3 + c_3 \leq 4000, a_4 + b_4 + c_4 \leq 1000, \\ P = 32.4A + 31.5B + 30.6C - 26.0a_1 - 30.6a_2 - 29.2a_3 - 29.8a_4 - 26.0b_1 - 30.6b_2 - 29.2b_3 - 29.8b_4 - 26.0c_1 - 30.6c_2 - 29.2c_3 - 29.8c_4]$$